Ivan Chen

(917) 703 9398

linkedin.com/in/anivanchen | New York, NY

ic8437@rit.edu

Highly motivated **Computer Engineering** student looking to leverage a **strong background in automation and robotics** at a **Spring/Summer 2027 Co-op**. Contributes technical proficiency in C/C++, VHDL, Python, systems design, physics-based simulation, and controls systems in fast-paced, cross-functional, and innovative engineering teams.

EDUCATION

Rochester Institute of Technology August 2024 – (Expected) December 2028

Bachelor of Science in Computer Engineering Technology | GPA: 3.65/4.0

Relevant Coursework: Microcontroller Systems, Digital Systems Design, Advanced Electronics**,** Circuits 1 & 2, APCSA

SKILLS

Technical Skills: C/C++, VHDL, MSP432, Python, Robotics, Computer Vision, Embedded Systems, Motion Planning, Sensors (LiDAR, Cameras, IMUs, Encoders, GPS), Control Systems, Systems Design, CAD, Statistics

Tools: Quartus Prime, ModelSim, Multisim, Oscilloscopes, OpenCV, Fusion360, Git, Docker, MS Excel, SQL

EXPERIENCE

Autonomous Software Engineer August 2024 – Present

FSAE Racing Performance Team @ RIT

- Develop a high-fidelity multi-threaded vehicle dynamics simulation in C using a bicycle model and torque vectoring to analyze and test a full perception, planning, and control pipeline.
- Engineer a 3D perception stack to process simulated LiDAR frames, sent via UDP, into point clouds with detection and classification algorithms like RANSAC ground plane removal and A-BIRCH for cone clustering.
- Implement a Stanley vehicle controller for trajectory tracking, migrating from previous pure pursuit approach.
- Create a baseline path planner calculating track centerline using Delaunay triangulation and cubic interpolation.
- Research advanced control, including LQR and MPC, and high-fidelity SLAM solutions using LIO-SAM.

Software Engineering Intern June 2025 – August 2025

NYC Mayor's Office of Management and Budget

- Automated end-to-end tax data pipelines with automated report generation/notification with MS Power Automate.
- Reduced daily data cleaning / processing time by 93% to 1.5 minutes, using Pandas and Python.
- Architected internal tax legislation research LLM + RAG system, with ChromaDB and LLaMA.

Systems Engineer August 2024 – June 2025

RIT Aero Design Team

- Developed C drivers for interfacing with inertial sensors, GPS, and servo motors on control surfaces.
- Implemented a high-level finite state machine to move between control states such as turning, landing, etc.
- Applied Unscented Kalman Filter (UKF) with sensor data and PID control for autonomous glider navigation.

Vice President of Software Engineering October 2020 – June 2024

FIRST Robotics, StuyPulse

- Developed a multi-camera localization system using AprilTags, ArUco, and OpenCV on headless Linux coprocessors (Raspberry Pi, Orange Pi) with 0.5-inch accuracy.
- Implemented and tested Java code using WPILib for 8 high-performing robots, enabling complex feature-sets.
- Engineered control systems for complex mechanisms using state machines, FF/PID controls, and localization.
- Led 50+ and taught over 200+ members in software and robotics topics and managing interdepartmental relations.
- Perform root cause analysis on software/hardware issues for 48+ teams; presented workshops to 400+ attendees.

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EDUCATION

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Stuyvesant High School September 2020 – June 2024
Advanced Regents Diploma with Honors in Mathematics and Science

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Technical Skills: C/C++, VHDL, MSP432, Python, Robotics, Computer Vision, Embedded Systems, Motion Planning, Sensors (LiDAR, Cameras, IMUs, Encoders, GPS), Control Systems, Systems Design, CAD, Statistics

Tools: Quartus Prime, ModelSim, Multisim, Oscilloscopes, Altium, OpenCV, Fusion360, Git, Docker, MS Excel, SQL

EXPERIENCE

Autonomous Software Engineer August 2024 – Present
FSAE Racing Performance Team @ RIT

- Develop a high-fidelity multi-threaded HIL vehicle dynamics simulation in C using a bicycle model and torque vectoring to analyze and test a full perception, planning, and control pipeline.
- Engineer a 3D perception stack to process simulated LiDAR frames, sent via UDP, into point clouds with detection and classification algorithms like RANSAC ground plane removal and A-BIRCH for cone clustering.
- Implement a Stanley vehicle controller for trajectory tracking, migrating from previous pure pursuit approach.
- Create a baseline path planner calculating track centerline using Delaunay triangulation and cubic interpolation.
- Research advanced control, including LQR and MPC, and high-fidelity SLAM solutions using LIO-SAM.

Software Engineering Intern June 2025 – August 2025
NYC Mayor's Office of Management and Budget, Income Unit

- Automated end-to-end tax data pipelines with automated report generation/notification with MS Power Automate.
- Reduced daily data cleaning / processing time by 93% to 1.5 minutes, using Pandas and Python.
- Architected internal tax legislation research LLM + RAG system, with ChromaDB and LLaMA.

Systems Engineer August 2024 – June 2025
RIT Aero Design Team

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- Led 50+ and taught over 200+ members in software and robotics topics and managing interdepartmental relations.
- Perform data driven root cause analysis on software/hardware issues for 48+ teams and presented workshops to 400+ attendees.

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EDUCATION

Rochester Institute of Technology August 2024 – (Expected) December 2028

Bachelor of Science in Computer Engineering Technology; GPA: 3.85/4.0

Stuyvesant High School September 2020 – June 2024

Advanced Regents Diploma with Honors in Mathematics and Science

Relevant Coursework: Microcontroller Systems, Digital Systems Design, Electronic Devices**,** Circuits 1 & 2, APCSA

SKILLS

Technical Skills: Robotics, Computer Vision, Embedded Systems, Motion Planning, Classical Controls, CAD, Systems Design, Data Pipelines, RESTful APIs, Computer Networking, Statistical Analysis

Languages: C/C++, VHDL, Python, Java, JavaScript (React, Node.js, Express.js), SQL, Bash, GitHub Actions

Tools: Fusion360, Altium, Quartus Prime, Git, Docker, Vercel, AWS, Ollama, Proxmox, Pico SDK, MS Excel

EXPERIENCE

Autonomous Vehicle Engineer August 2024 – Present

FSAE Racing Performance Team @ RIT

- Research and develop localization, vision, and planning algorithms to achieve end-goal of autonomous racing.
- Develop cone detection and pairing with simulated LiDAR data using DBSCAN clustering and ground filtering.
- Design and test a collision impact testing rig with custom circuitry for precise data acquisition and analysis.

Data Engineering Intern June 2025 – August 2025

NYC Mayor's Office of Management and Budget, Income Unit

- Automated end-to-end tax data pipelines with automated report generation/notification with MS Power Automate.
- Reduced daily data cleaning / processing time by 93% to 1.5 minutes, using Pandas and Python.
- Architected internal tax legislation research LLM + RAG system, with ChromaDB and LLaMA.

Systems & Algorithms Engineer August 2024 – June 2025

RIT Aero Design Team

- Developed C software for interfacing with inertial sensors, GPS, and servo motors on control surfaces.
- Implemented a high-level finite state machine to move between control states such as turning, landing, etc.
- Applied Unscented Kalman Filter (UKF) with sensor data and PID control for autonomous glider navigation.

Senior Software Engineer October 2020 – June 2024

FIRST Robotics Competition Team 694, StuyPulse

- Implemented a multi-camera localization system using AprilTags, ArUco, and OpenCV with 0.5-inch accuracy.
- Developed and tested Java code using WPILib for 8 high-performing robots, enabling complex feature-sets.
- Engineered control systems for complex mechanisms using state machines, FF/PID controls, and localization.
- Led 50+ and taught over 200+ members in software and robotics topics and managing interdepartmental relations.
- Debugged critical software/hardware issues for 48+ teams and presented workshops to 400+ attendees.

Lead Web Developer November 2021 – June 2024

The Stuyvesant Spectator

- Redesigned stuyspec.com using Next.js and MongoDB, reducing document-heavy load times to 400ms.
- Moved web hosting from AWS Lightsail to Vercel, eradicating downtime.
- Reworked the Content Management System for better efficiency and user experience.

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Relevant Coursework: Microcontroller Systems, Digital Systems Design, Low Level Programming, APCSA

SKILLS

Technical Skills: Systems Design, Robotics, Computer Vision, Embedded Systems, Data Pipelines, Statistical Analysis, Computer Networking, Classical Controls, CAD, Technical Writing, Research, Project Management

Languages: C/C++, Python, Java, JavaScript (React, Node.js), SQL, Bash

Tools: Git, Docker, Vercel, Ollama, Pico SDK, AWS, Azure DevOps, MongoDB, Pandas, Matplotlib, Numpy

EXPERIENCE

Data Engineering Intern June 2025 – August 2025
NYC Mayor's Office of Management and Budget, Income Unit

- Automated end-to-end tax data pipelines with automated report generation/notification with MS Power Automate.
- Reduced daily data cleaning / processing time by 93% to 1.5 minutes, using Pandas and Python.
- Architected internal tax legislation research LLM + RAG system, with ChromaDB and LLaMA.

Electronics Engineer August 2024 – Present
FSAE Racing Performance Team @ RIT

- Design and test a collision impact testing rig with custom circuitry for precise data acquisition and analysis.
- Execute debugging and repairing complex electronics systems on the F30 electric race car.
- Research and develop localization, vision, and planning algorithms to achieve end-goal of autonomous racing.

Controls & Firmware Engineer August 2024 – June 2025
RIT Aero Design Team

- Developed low-level C firmware for interfacing with inertial sensors, GPS, and servo motors on control surfaces.
- Implemented a high-level finite state machine to move between control states such as turning, landing, etc.
- Applied Unscented Kalman Filter (UKF) with sensor data and PID control for autonomous glider navigation.

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- Engineered control systems for complex mechanisms using state machines, FF/PID controls, and localization.
- Debugged critical software/hardware issues for 48+ teams and presented workshops to 400+ attendees.

Vice President of Software Engineering

- Led a team of 50+ software engineers within a 150+ member organization, overseeing software development.
- Strategically directed SWE efforts at regional and national competitions, resulting in top-tier placements.
- Coordinated with mechanical engineers to diagnose and resolve issues, ensuring adherence to strict timelines.

Senior Software Engineer

- Developed & tested Java code using WPILib for three highly ranked robots, Excelsior, Jim, and Izzi, and 5 others.
- Maintained StuyLib, a library for controls algorithms, signal processing, and Ethernet-based communication.
- Engineered control systems using state machines and feedforward/feedback systems for mechanisms including a double-jointed arm, telescoping mechaism, a color-sorted conveyor system, and automated self-driving systems.
- Implemented a multi-camera localization and calibration system using AprilTags, ArUco, and OpenCV.
- Applied UKF to for data fusion with wheel odometry and perform autonomous driving and scoring routines.
- Mentored 7 teams in Java programming and presented to 400+ people over 3 years on advanced software topics.
- Provided critical debugging aid to 48+ teams around the world in complex software and hardware issues.

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EDUCATION

Rochester Institute of Technology August 2024 – Present

- *College of Engineering Technology*: Bachelor of Science in Computer Engineering Technology, *GPA*: **4.0/4.0**

Stuyvesant High School September 2020 – June 2024

- Advanced Regents Diploma with Honors in Mathematics and Science
- *SAT*: Math **750** | Reading & Writing **740**

Relevant Coursework: Microelectronics Systems, Low Level Programming, AP CS A, AP Statistics, Analog Electronics

SKILLS

Programming Technologies: C/C++, Java, Python, JavaScript, SQL, MongoDB, Docker, Git

Technical Skills: Robotics, Computer Networking, Circuits & CMOS transistors, Classical Controls Engineering, Electronics Design, Soldering, Computer Vision, Statistical Analysis, CAD, Quartus Prime

Other Skills: Microsoft Office, Technical Writing, Project Management, Research Skills

EXPERIENCE

RIT Aero Design Team, *Controls & Firmware Engineer* August 2024 – Present

- Write low-level C programs for interfacing with inertial sensors, GPS, and servo motors on control surfaces.
- Utilize sensor data in Unscented Kalman Filter (UKF) and PID control for autonomous glider navigation.

RIT FSAE Racing Performance Team, *Electronics Engineer* August 2024 – Present

- Build PCBs with surface and through hole components and firmware programming using C.
- Debug and repair electronics issues on F30, a previous electric race car.

FIRST Robotics Competition Team 694 October 2020 – June 2024

Vice President of Software Engineering

- Led a team of 150+ members, including 50+ software engineers, overseeing software development.
- Organized and directed SWE efforts at regional and national competitions, resulting in top-tier placements.
- Collaborated with mechanical engineers to identify and resolve issues, along with organizing timelines.

Senior Software Engineer

- Developed & tested Java code using WPILib for three highly ranked robots, Excelsior, Jim, and Izzi, and 5 others.
- Maintained StuyLib, a library for controls algorithms, signal processing, and Ethernet-based communication.
- Devised code to control a novel drivetrain solution, swerve drive, with autonomous driving functionality.
- Engineered control systems using state machines and feedforward/feedback systems for mechanisms including a double-jointed arm, telescoping mechaism, a color-sorted conveyor system, and automated self-driving systems.
- Implemented a multi-camera localization and calibration system using AprilTags, ArUco, and OpenCV.
- Utilized UKF to merge data with wheel odometry and perform autonomous driving and scoring routines.
- Mentored 7 teams in Java programming and presented to 400+ people over 3 years on advanced software topics.
- Provided aid to 48+ teams around the world in debugging software and hardware issues.

The Stuyvesant Spectator, *Lead Web Developer* November 2021 – June 2024

- Completely redesigned and built a new website, stuyspec.com, using Next.js, reducing load times to 400ms.
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Relevant Coursework: Microeconomics**,** AP Statistics, Microcontroller Systems, Low Level Programming, AP CS A

SKILLS

Programming Technologies: C/C++, Java, Python, JavaScript, SQL, MongoDB, Docker, Git

Technical Skills: Software Design, Data Cleaning/Parsing, Robotics, Computer Networking, Circuits & CMOS transistors, Classical Controls Engineering, Electronics Design, Soldering, Computer Vision, CAD, Quartus Prime

Other Skills: Project Management, Statistical Analysis, Research Skills, Microsoft Office, Technical Writing

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